

TRANSPORT FINDINGS

Right-wing Votes Relate to Delays in Bicycle Network Development

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Findings

Paris has implemented only 43% of its 2021-2026 bicycle network plan by its 2026 deadline, but the delays are not equally distributed. Correlating political voting and bicycle infrastructure data, we find that boroughs with a higher share of right-wing votes are also boroughs with the largest delays in bicycle infrastructure development. Although the number of voting units is only 17, this correlation is remarkably strong and significant. At the city scale, for each additional 1% of right-wing votes, we find 0.86% less protected bicycle lanes built.

1. Questions

Urban bicycle network development is more a political than a technical problem (Buehler 2021). Political opposition is common in motonormative societies (Walker et al. 2022; Henderson and Gulrud 2019; Wild et al. 2018), manifested in systemic local obstructions of infrastructure development causing functional collapse on the network level (Szell et al. 2022) and stifling cycling uptake with its numerous societal benefits (Schön et al. 2024; Fosgerau et al. 2023; Gössling et al. 2019). Therefore, the political reasons for such delays must be scrutinized, to identify and fix the practical bottlenecks of bicycle network development.

Paris is a good example of a city dedicated to improve its bicycle infrastructure (Moran 2022; Liu et al. 2026). The modern push towards more cycling in France started from environmental concerns in 1994 via the creation of the *Comité de suivi de la politique du vélo*, during Paris' right-wing governments of Chirac, then Tiberi, which was then more strongly supported by successive left-wing governments, first under Delanoë, then Hidalgo under the banner of the *15-minute city* (Dauncey 2012; Narangoda et al. 2024; Moreno 2024). Although up to March 2026, Hidalgo was the city's socialist mayor, the city's different *arrondissements* (local administrative unit corresponding to a metropolitan borough) may be governed by arrondissement mayors and councils of different political colors.

Here we focus on the "Paris Plan Vélo 2021-2026" (Municipality of Paris 2021; Narangoda et al. 2024), a 182 km plan of protected bicycle lanes. Only 43% of the planned infrastructure has been constructed city-wide by its 2026 deadline (Paris en Selle 2026). While locally elected councils have in theory only a consulting role (Code général des collectivités territoriales 2026), public discussion has found delays and the overall bikeability to differ

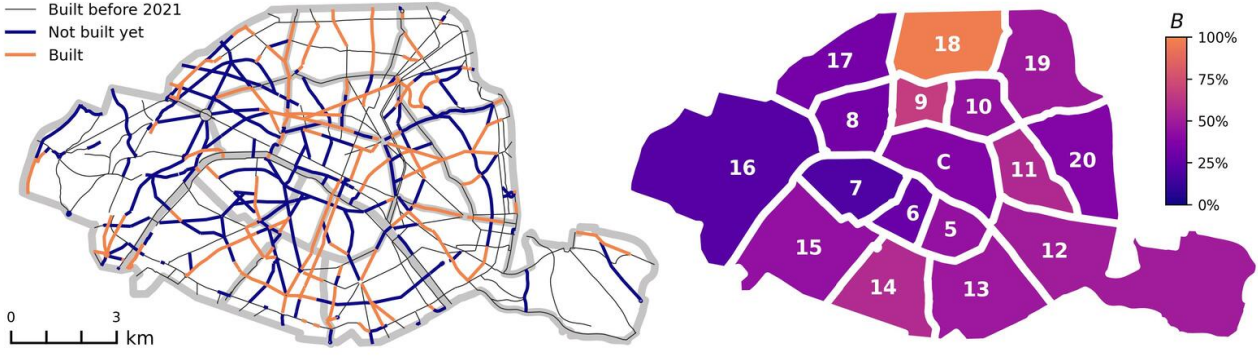


Figure 1. Left: Paris' Plan Vélo 2021-2026 progress on street level, Right: share of bicycle lanes built B aggregated by political voting unit.

between boroughs depending on the local elected official (Paris en Selle 2023). Such discussions motivate us to investigate whether there is a sound statistical relationship between Paris' voting patterns and the development progress of its bicycle network.

2. Methods

We measure the progress of the bicycle network plan using the latest street-level data, collected by the local cycling advocacy organization (Paris en Selle 2026). We aggregate the data by voting units, see below, and call this variable the share of bicycle lanes built B , see [Figure 1](#).

Voting results come from Paris Data (2020) on the municipal election of 2020, where citizens voted for their local council. Each of the 20 councils correspond to an arrondissement. The first four boroughs are aggregated in a single unit named “Centre” (denoted as “C”), leaving us with 17 voting units. We use the results of the first voting round as they best capture the voting intent of citizens, without voting system artifacts like second-round list exclusions. We classify lists using the authoritative nuance list of the Ministère de l'Intérieur (2020), subsuming the party lists LEXG, LFI, LDVG, LUG, LVEC, and LDIV as left and LUD, LDVD, LRN as right; the central lists LUC and LDVC do not belong to either class.¹

We also use the median income Y (INSEE 2024c) and the approximated mean age A (INSEE 2024b) normalized to the range 0-1, and cycling/driving commuter shares M_C and M_D (INSEE 2024a), to investigate additional relationships. As the age data comes in age brackets we can calculate an approximation of the mean, but not a median age.

¹ LEXG: Extrême gauche, LFI: La France insoumise, LDVG: Divers gauche, LUG: Union de la gauche, LVEC: Europe Ecologie-Les Verts, LDIV: Divers | LUD: Union de la droite, LDVD: Divers droite, LRN: Rassemblement National | LUC: Union du centre, LDVC: Divers centre

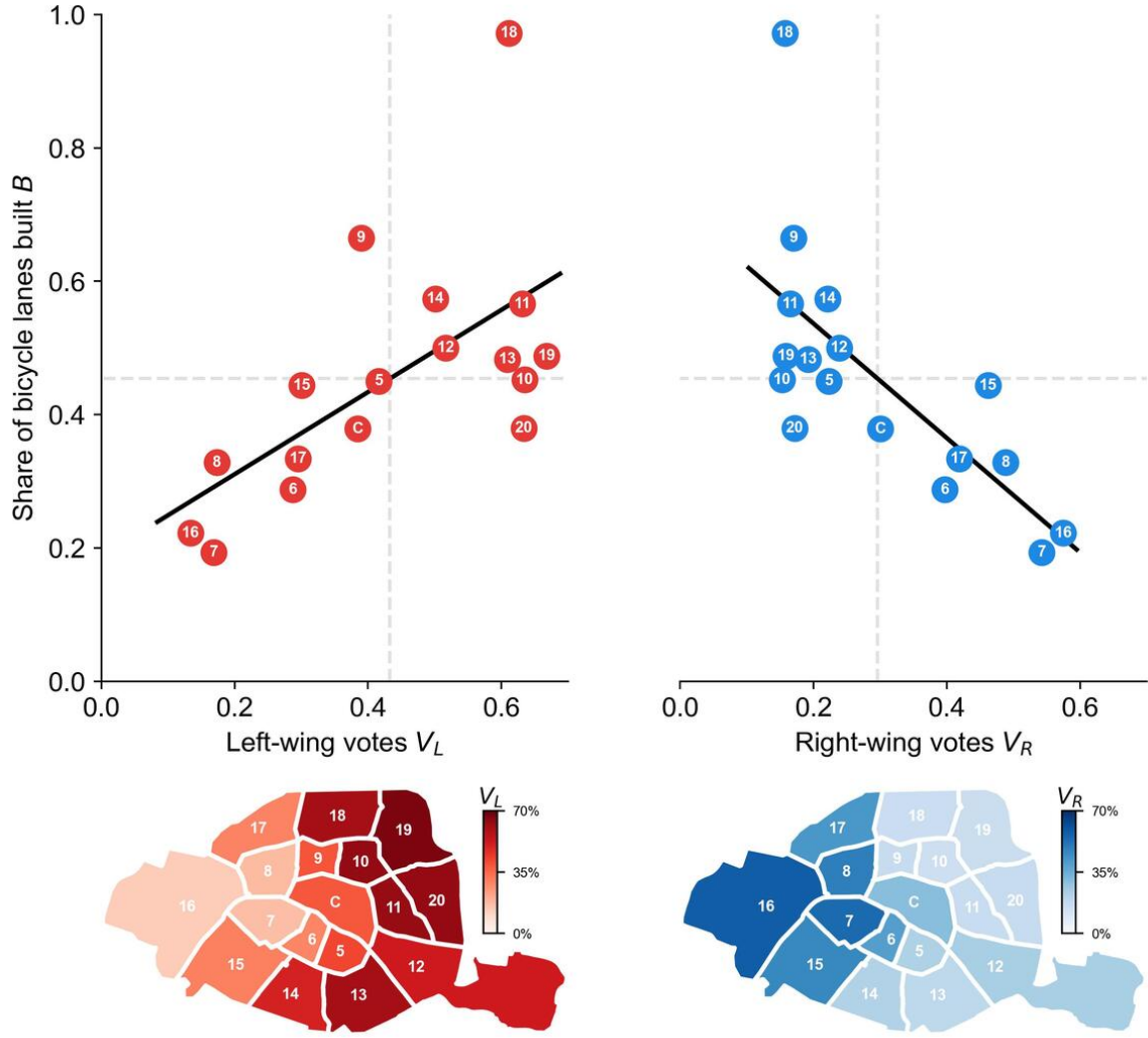


Figure 2. Left: Scatterplot of the share of bicycle lanes built B against left-wing votes V_L , Right: against right-wing V_R votes, with corresponding maps of the voting patterns. The black lines show fitted linear regressions. Dashed gray lines show the average values for each axis.

We make publicly available all code and processed data in the following repository: https://github.com/csebastiao/paris_vote_bicycle

3. Findings

We correlate built bicycle lanes B both with the votes of left and right political sides V_L and V_R and not with only one unitary left-right measure, to keep the analysis simple and to respect the inherent asymmetry due to the center parties. Further, it is worth to report different statistical indicators depending on the political side to explore potential asymmetries between opposition and support, in light of relevant psychological biases like loss aversion and status quo biases (Walker et al. 2022; Timmons et al. 2023; Kahneman and Tversky 1984).

We find a statistically significant positive correlation between left-wing votes V_L and built bicycle lanes B (Pearson $\rho = 0.62$, $p = 0.0081$), and an even stronger negative correlation between right-wing votes V_R and B ($\rho = -0.71$, $p = 0.0016$), see [Figure 2](#). The slope of $\beta = -0.86$ for the latter implies that, on average, for each additional percent of right-wing votes, 0.86% less protected bicycle lanes are built. The significances of these correlations are remarkable considering the low $n = 17$. Borough 18 is an outlier, which had the least bicycle infrastructure planned (2.4% of the entire plan for 5.6% of the Paris area) of which it implemented 97%.

Despite the clarity of these results, they need to be interpreted with care: Correlation is not causation, small sample size makes statistical results less reliable, and there are many potential confounders possible. To this end, we explore the additional demographic variables introduced in the Methods, see [Table 1](#). Of these, the negative correlations between B and the median income Y ($\rho = -0.61$), the mean age A ($\rho = -0.67$), and the share of commuter driving M_D ($\rho = -0.62$) are significant, all at $p < 0.01$. Furthermore, all variables correlate among themselves. Trivially, V_L correlates negatively with V_R ($\rho = -0.93$, $p = 5.4 \times 10^{-8}$) as does M_C with M_D ($\rho = -0.76$, $p = 0.00046$). Right-wing votes V_R correlate positively with Y ($\rho = 0.80$, $p = 0.00012$), A ($\rho = 0.69$, $p = 0.0023$), and M_D ($\rho = 0.81$, $p = 0.000082$). Left-wing votes V_L correlate negatively with Y ($\rho = -0.90$, $p = 1.1 \times 10^{-6}$) and A ($\rho = -0.61$, $p = 0.0090$), and positively with M_C ($\rho = 0.59$, $p = 0.012$), in line with R  rat and Ravalet (2025). A correlates also positively with Y ($\rho = 0.70$, $p = 0.0019$) and M_D . ($\rho = 0.65$, $p = 0.0050$) In summary, driving, higher mean age, right-wing votes, and bicycle network delays are more strongly related than cycling, lower mean age, left-wing votes, and bicycle network development.

To verify that the votes data V_R carries relevant information beyond the demographic variables, we perform a hierarchical regression, comparing a multilinear regression model predicting B with all demographics variables (Y , A , M_D , and M_C) to a model which additionally includes the independent variable V_R . Both models are significant ($p < 0.05$) but each individual variable is not significant due to strong multicollinearity. Nevertheless, adding V_R does increase significantly the model's R^2 ($\Delta R^2 = 0.031$, $p = 0.017$) and its adjusted R^2 ($\Delta R^2_{\text{adj}} = 0.021$). Although this result has to be interpreted carefully due to the few data points, it highlights the statistical relevance of the voting patterns even after controlling for demographics.

We also find a considerable positive spatial autocorrelation. Computing Moran's I with rook contiguity, we find $I(B) = 0.17$, $I(V_L) = 0.64$, $I(V_R) = 0.61$, $I(Y) = 0.43$, $I(A) = 0.43$, $I(M_C) = 0.36$, $I(M_D) = 0.38$, all with $p < 0.01$, and a general west-east trend, which calls for further care in the interpretation.

Table 1. Correlations and linear regressions between the share of bicycles lanes built B and each available variable, aggregated at the voting unit level ($n=17$). ** $p < 0.01$.

Share of bicycle lanes built B	Pearson correlation ρ	Slope β
Left-wing votes V_L	0.62**	0.62
Right-wing votes V_R	-0.71**	-0.86
Median income (normalized) Y	-0.61**	-0.38
Mean age (normalized) A	-0.67**	-0.42
Share of commuter cycling M_C	0.36	3.2
Share of commuter driving M_D	-0.62**	-2.7

Our correlational analysis is just a beginning which cannot consider cases of further possible confounders. For example, there could be a potential causal dependence between right-wing voting and boroughs with streets that are harder to adapt to cycling infrastructure, or cases of potential reverse causation such as car-centric boroughs attracting more right-wing inhabitants. Even if there was a causal relationship, one would need to consider counterfactuals: Instead of developing bicycle infrastructure, could the local council have generated other benefits?

Bicycle infrastructure development is a complex process with nontransparent political responsibilities, incentives, and power structures, divided among a diverse network of actors that requires considerable efforts to untangle (Lavin 2025; Elvarsson et al. 2026). For example, while in Paris the mayor can set up city-wide cycling infrastructure plans, depending on the kind of infrastructure planned and the type of street, local arrondissement councils and other actors like the police or the *Architectes des Bâtiments de France* have to be consulted and can influence bicycle infrastructure development. Due to the nontransparency of the underlying processes, which are important to unveil in future research, the extent of this power is unclear.

We hypothesize a phenomenon of local obstruction, either through direct obstruction by the local arrondissement council, or through indirect effects like the city mayor targeting areas where local support is stronger. If true, this “local obstruction hypothesis” could give a causal reason for the correlations we found between votes and infrastructure delays.

Anne Hidalgo, the left-leaning mayor of Paris from 2014 to 2026, explicitly stated her objective to “exit car-dependency” in the Plan Vélo 2021-2026 (Municipality of Paris 2021), a plan criticized by Rachida Dati, Hidalgo’s right-leaning municipal opponent and mayor of the 7th borough, as it would “paralyze Paris” (BFMTV 2020). Given this ongoing polarized discussion, together with the literature reporting local obstructions elsewhere (Henderson and Gulsrud 2019; Wild et al. 2018), we interpret our result as a reasonable piece of evidence for the local obstruction hypothesis, which we call for exploring further towards a causal disentangling of decision mechanisms (Elvarsson et al. 2026) and potentially underlying psychological

effects (Walker et al. 2022; Kahneman and Tversky 1984; Timmons et al. 2023). If the hypothesis is true, it might turn out counterproductive for right-wing parties to oppose new bicycle infrastructure, given the non-negligible proportion of cyclists among their supporters (Rérat and Ravalet 2025). In any case, for many reasons including environment and public health, we must better understand and fix the bottlenecks in the urgently needed development of bicycle infrastructure (Brand et al. 2021; Miner et al. 2024).

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